

The Process of Constructing *Postcard from Sertões* within a Csound Environment - ICSC 2026

Catarina Schmitt Rossi¹ and Bruno Gazoni²

^{1 2} Arts Institute of the University of Campinas IA-UNICAMP
c116467@dac.unicamp.br
b116306@g.unicamp.br

Abstract. This paper describes the creative and technical process of the performance *Postcard from Sertões*. The work details the transposition of Almeida Prado's *Sertões* (1978) for solo viola [1] into the interactive electroacoustic domain, inspired by the compositional economy of James Tenney's *Postal Pieces* (1971) [2]. The research was developed within an Interpretive Experimentation Laboratory, which consisted of distilling Prado's "transtonal system" into four essential sound blocks, interpreted as "clangs", primary perceptual units based on Tenney's theory. The performance establishes a bridge between written tradition and orality by distributing musical materials between two distinct performers. An acoustic interpreter plays the Brazilian *rabeca* and the viola, while an algorithm interpreter manipulates the audio input in real-time. The implementation, done via Csound and Cabbage, involves a digital signal processing (DSP) system operated by a postcard-inspired visual interface. Through four looper tracks enhanced by effects, the algorithm interpreter reshapes the clangs' identities. The laboratory explored and refined the interaction protocol between the musicians through remote testing and in-person rehearsals. The artistic practice suggests that technological mediation promotes a transformation of the work, resulting in a new hybrid composition that functions as a "virtual postcard", reactivating the symbolic resonances of the *sertão* through a phenomenological and immersive listening zone.

Keywords: Csound, Cabbage, Almeida Prado, James Tenney, Performance, Viola, Rabeca.

1 Introduction

This investigation analyzes the convergence between instrumental idiomatism and digital mediation, centering on the composition *Sertões* (1978) by Almeida Prado [1]. This performance proposal seeks to recontextualize elements of the original piece for solo viola through the lens of the phenomenology of sound, employing methodological elements from the work of James Tenney [2] [3]. Furthermore, it utilizes real-time digital signal processing (DSP) via Csound to implement the resonance field inherent to the composers' writing.

Sertões represents Almeida Prado's only foray into the repertoire for solo viola and can be viewed as an exemplar of cultural hybridism, described by Canclini as "sociocultural processes in which discrete structures or practices, which existed separately, combine to generate new structures, objects, and practices" (trans. ours.) [4]. The *sertão* refers to the vast, semi-arid backlands of Northeastern Brazil. In this performance, it is treated as a "symbolic field", a collection of values and territorial imaginaries expressed through specific timbral and gestural choices [5]. This territory is sonically embodied by the *rabeca* (a Brazilian folk fiddle), whose cantabile style and use of open-string pedal notes evoke the atmosphere and oral traditions of this region [6].

Prado's timbral poetics is based on a "transtonal system", in which form emerges from the organization of sound matter and gesture [7]. For this analysis, we refer to the concepts of "neocadences" and "auditory redundancies," where the insistent repetition of ambiguous tonal centers establishes what the author calls "referential sonority" [8].

In dialogue with Prado's work, this project incorporates the aesthetics of James Tenney, specifically his *Postal Pieces* cycle (1954–1971). As described by Larry Polansky, these works consist of ten minimalist musical scores originally written on postcards and sent to friends and colleagues, representing a radical economy of means that strips musical materials to their "bare bones" to focus the

listener's attention on specific perceptual phenomena [9]. Among the most recognized works in this cycle are *Having Never Written a Note for Percussion*, *KOAN* (for solo violin), and *(night)* [10]. Tenney's approach is fundamentally guided by the phenomenology of sound explored in his treatise *Meta+Hodos* [3], which explores the perception of coherent units of sound that emerge through listening with attention to variability, continuity and context [11].

Within this poetic framework, listening becomes the structuring axis, emphasizing the perception of sound masses and densities over traditional thematic developments [11]. The formal organization in Tenney's work seeks a state of suspension where temporality is mediated by minimal variations in spectral parameters, closely mirroring the "mobile" sonic quality that Almeida Prado sought for his compositions, a state where the music "moves while standing still". By distilling Prado's dense "transtonal system" into these primary perceptual units, the performance creates a hybrid zone of listening where the minimalist directives of the postcards activate the deep symbolic resonances of the *sertão*.

This proposal presents digital processing as an interpretive agent, where the technical configuration redefines the interaction between the instrumentalist and the sound, organized through signal extraction and electronic manipulation. Thus, a second musician acts as an "algorithm interpreter," responsible for controlling parameters within the digital interface in real-time.

For this purpose, the central objective of this article is to analyze the performance of *Postcard from Sertões*¹ as an instance of meaning production mediated by digital tools, investigating how DSP can enhance the concepts of hybridism and sonic phenomenology. To this end, the text is subdivided into sections detailing performance-oriented analysis, algorithmic implementation procedures in Csound, and, finally, the results of the interpretive experimentation laboratory, consolidating the role of the digital as a co-creator of the contemporary soundscape.

2 Theoretical Foundation

The performance *Postcard from Sertões* is anchored in a zone of convergence between Almeida Prado's "transtonal system" and James Tenney's phenomenology of sound. Both authors, although originating from distinct contexts, share an interest in auditory perception and the organization of resonance as the structural axes of musical form.

Transtonalism is a language consciously developed by Almeida Prado starting in the 1970s, based on the systematization of principles related to the organization of resonance and timbral exploration, freely utilizing harmonics and lower resonances to manipulate sound [12]. The primary characteristic of this system is the proposition of an ambiguous tonal centrality through "neocadences", procedures that employ repetition and auditory redundancy as a means of musical memorization and comprehension [7].

Within this poetic framework, form is articulated through sound blocks, which function as fixed or static harmonies intended to generate specific "colors" in the work, also referred to as "referential sonorities" [8]. In *Sertões*, the first sound block is established by the persistence of the open G string, acting as a pedal point and creating a static resonance field that serves as a foundation for the melody in double stops.

In dialogue with the density of Prado's work, the proposal incorporates the concepts of James Tenney, who seeks to place the direct experience of listening at the center of music theory [11]. Tenney proposes a perceptual classification of sound configurations, favoring the concept of the clang over terms such as "motive" or "referential sonority". The clang is defined as a primary musical unit, a "singular auditory Gestalt" that imposes itself on perception through its internal cohesion and segregation from the surrounding acoustic context [11]. For Tenney, musical form is not a predetermined schema, but rather the perception of differences and changes within these units. In his *Postal Pieces*, Tenney highlights the formation of these units through deliberately limited directives that nonetheless possess a high gestural potential [10].

Thus, the project promotes an "ecology of ideas" wherein Prado's "transtonal system" is presented through Tenney's lens. The creative process involved distilling the original structure of *Sertões* into

¹ Performance video available at: https://youtu.be/qSZzC_Um2cQ?si=7irRTJOxUVdlcl7v

four fundamental sound blocks, which were reinterpreted as clangs. This structural simplification allows Prado's redundancy to converge with Tenney's economy, transforming the original piece into a sonic travelogue.

3 Technical Implementation

In this section, we detail the Digital Signal Processing (DSP) architecture developed in the Csound environment and the graphical user interface implemented via the Cabbage framework. The Csound patch was designed to process elements of the performance in line with Almeida Prado's poetics, organizing and overlaying sound blocks (or clangs, in Tenney's framing) as multiple coexisting textures. As they re-create and re-tread a selection of sonic memories from the original piece, the performer stands as the primary generator of the gestures that compose the sound field, legitimizing them as an active agent in co-creation and the production of meaning [5].

The plugin was developed with VST exporting in mind, so global configuration bypasses native audio/MIDI drivers and operates with a control rate buffer of $\text{ksmps} = 32$. The patch architecture (see flowchart in Appendix 1) is built upon a custom opcode named *LooperEngine*, which manages memory allocation, state-machine logic (Empty/Recording/Playing), and playback phase. Memory buffers are initialized at startup via *ftgen* as empty arrays of 60 seconds, the stipulated maximum loop length. Playback is driven by a phasor opcode, whose frequency is calculated dynamically ($\text{sr}/\text{kLoopSamples}$) and scaled by a UI speed parameter (ranging from -2 to 2), permitting variable-speed and reverse playback.

Four distinct clangs were selected from the original piece to serve as the foundational textures. Each features a dedicated *LooperEngine* instance and feedback effects designed to transform the captured *rabeca* loops without affecting the live bypass signal.

a) Low Drone (Bordão): Sub-bass synthesis and filtering

Aiming to reinforce low end presence, this track utilizes a Phase Vocoder (PVS) network via the Short-Time Fourier Transform (STFT). The loop is analyzed using *pvsanal* (1024-sample FFT size and window, 256-sample overlap, Hanning window). The spectral data is duplicated and shifted down by one and two octaves using *pvscale*. The bins are recombined via *pvmix* and subjected to a 1.0-second spectral blur (*pvsblur*) to smear transients, making the low loop act more as a cushion to the upper parts. After resynthesis to the time domain (*pvsynth*), the signal is soft-clipped via a hyperbolic tangent function (*tanh*) with a 1.5x gain multiplier and routed through the *moogladder* low-pass filter opcode (UI cutoff range: 50–1500 Hz).

b) Melodies (Melodia): Spectral frequency shifting

The feedback effect idea for the melody is to recharacterize its timbre via absolute pitch shifting, acting as a metallic flavor over the original loop. Following *pvsanal* analysis, a moderate 0.4-second *pvsblur* smears the melodic contour. The *pvshift* opcode is then used to add a fixed, linear Hertz value to all frequency bins (UI range: 10–500 Hz), disrupting harmonic relationships before time-domain resynthesis.

c) Natural Harmonics (H. Naturais): Gated ping-pong shimmer

Natural harmonics are peppered through the stereo field as high end droplets. This loop is spectrally pitched up one octave (*pvscale* at 2.0), blurred (0.2s), and resynthesized. A tone low-pass filter (8000 Hz) removes piercing digital artifacts. To create rhythmic bursting, a stochastic gate evaluates a random sample-and-hold generator (*randomh*, operating at 8 Hz, outputting 0.0–1.0) against a UI "density" threshold (0.0–1.0). The resulting binary trigger is smoothed over 50ms using the *portk* opcode to prevent audio clicks. These gated bursts feed two independent delay lines (*delayr* with a 2.0-second memory space), tapped via cubic interpolation (*deltap3*) at 300ms (Left) and 600ms (Right). A UI width parameter (0.0–1.0) mathematically crossfades between the mono *aFxMono* sum and the hard-panned ping-pong signals.

d) Artificial Harmonics (H. Artificiais): Distortion and reverb wash

This material will have its presence reinforced by distortion and reverb. The loop is pushed into a tanh soft-clipper scaled by a UI drive parameter (ranging from 1 to 50, compensated by a 0.4x volume multiplier). It is then sent through a freeverb algorithmic reverberator (room size: 0.8, high-frequency dampening: 0.2). To prevent the broadband energy of the distortion from overwhelming the master bus, the stereo field is downmixed to mono and passed through a steep 150 Hz Butterworth high-pass filter (butterhp).

e) Ending Global Processing

The piece is conducted towards an ending via global loop offset manipulation, making all sequences shorter, and master reverb to fade out the definite attacks and sound identities into a blur. The global DSP chain allows the operator to synchronously modify the phase boundaries of all four discrete loops. Master start/end offsets (ranging from -1.0 to 1.0) are summed with local loop bounds and constrained via the limit opcode, allowing the operator to collapse phrases into tight, fragmentary stutters. The master "Running" toggle acts as a global gate; when toggled off, it dampens playback over 50ms (via portk). Crucially, this gated audio is routed into a Feedback Delay Network (FDN) reverberator (reverbrc with 0.95 feedback and 12 kHz cutoff) after the gate, allowing the accumulated master reverb tail to organically ring out into silence.

Keeping with the theme of Tenney's Postal Pieces, the Cabbage graphical interface, configured as an event-driven GUI (`guiMode("queue")`), functions as a virtual postcard. It combines succinct operational directives with generic musical notation that reflects the chosen clangs without quoting them directly. Lacking clefs, time signatures, or tempo markings, the UI serves simultaneously as an evocative prompt for musical memory and the functional control surface for the Csound architecture.

4 The Interpretive Experimentation Laboratory

The development of Postcard from Sertões was conducted within an Interpretive Experimentation Laboratory, a collaborative environment where artistic research and technical programming informed one another through an iterative process. This laboratory was essential for testing, adjusting, and consolidating the interaction strategies implemented in Csound.

The first stage involved an analysis of Almeida Prado's Sertões score, alongside the study of James Tenney's Postal Pieces. Simultaneously, the DSP architecture was programmed in Csound, focusing on real-time signal manipulation. To bridge the gap between the acoustic and digital domains, the acoustic interpreter recorded the work for solo viola and sent the files to the algorithm interpreter for remote testing, ensuring the initial stability of the DSP chain.

During three refinement sessions, the performers reached a consensus on the work's formal structure, isolating four clangs from the original composition, poetically interpreted as "memories". These laboratory sessions were based on experimentation and artistic research, seeking to find gestures and methods consistent with the expectations of the performers. For instance, initial attempts to process the viola's harmonic sections through heavy distortion and reverberation resulted in a muddy texture that completely obscured Prado's "transtonal" centers. Consequently, this approach was rejected.

The decision to open the piece with the *rabeca* aims to incorporate elements that align with the intrinsic language of the work. The use of double stops, the title, and the character of the initial theme suggest the composer drew direct inspiration from this musical universe. Simultaneously, presenting the remaining clangs on the viola, the instrument for which the original work was composed, seeks to establish a dialogue between the written score and its oral reinterpretation, thereby allowing the digital signal processing (DSP) to enhance the instrument's natural resonance during the more complex harmonic sections.

The transition between clangs also presented challenges. The initial iterations resulted in somewhat chaotic textures, as establishing the necessary synchrony between the two musicians to initiate the gesture required practical refinement. This misalignment frequently created loops that started mid-phrase and disrupted the musical sense. This lack of temporal cohesion was resolved during in-person rehearsals by establishing a communication protocol: the acoustic interpreter would provide

a clear visual cue and wait before moving to the next clang, ensuring the algorithm interpreter could accurately trigger the recording phase.

Another adjustment occurred in the visual domain. Initially, the Cabbage interface, designed to function as a virtual postcard, featured a blank background. Upon projecting it during rehearsals, the duo realized this visual absence disconnected from the immersive proposal. This was corrected by integrating photographs taken by the acoustic performer in the city of Caicó (RN), thereby adding the personal memories that inspired the performance.

The final performance format emerged from these reflections: the acoustic performer executes one sound block at a time following the score's sequence, while the algorithm interpreter captures these gestures, transforming them into looping structures and applying DSP effects. All laboratory sessions were documented through simultaneous video recordings of the instrumentalist and screen captures of the Cabbage interface, serving as the qualitative foundation for evaluating the performance's cohesion.

5 Results and Final Considerations

In conclusion, the project demonstrates the feasibility of integrating Brazilian oral traditions and erudite compositional systems into the domain of contemporary computer music. By filtering the density of Prado's "transtonal system" through James Tenney's "economy of means", the performance resulted in a sonic travelogue that preserves the phenomenological event. The process demonstrates the roles of the performer and the software operator as mediators who integrate the idiomatic characteristics of the instruments with the infinite possibilities of digital processing.

Thus, the tools developed in Csound enhanced the poetic dimension, encouraging an improvisation that reactivates musical memories, allowing *Sertões* to transition from a written record to a live electroacoustic practice, contributing to the continuity and expansion of Brazilian musical identity in the digital landscape. As final considerations, this work points to the need for further investigations into the transposition of traditional repertoires into interactive open-source environments.

References

1. A. Prado, *Sertões (1978)*, Darmstadt: Tonos 10349.
2. J. Tenney, *Postal Pieces*, 1971.
3. J. Tenney, "Meta + Hodos: a phenomenology of 20th-century musical materials and an approach to the study of form," *Frog Peak Music*, 1988.
4. N. G. Canclini, *Culturas Híbridas: estratégias para entrar e sair da modernidade*, Editora da Universidade de São Paulo, 2006.
5. C. V. d. O. Lemos, "Performance musical enquanto objeto de pesquisa," *Estudos Semióticos*, pp. 326-346, 2021.
6. C. S. Rossi, *Da cana de açúcar às mesas de som: histórias da rabeca através de rabequeiros*, Universidade Estadual de Campinas, 2019.
7. R. A. Nadai e R. d. C. Taddei, "O Sistema Transtonal em Metalosfera, de Almeida Prado," *MUSICA THEORICA*, pp. 215-230, 2016.
8. F. Scarduelli, *Khamaileon: fantasia para violão e orquestra de Almeida Prado*, Campinas, SP: Universidade Estadual de Campinas, 2009.
9. L. Polansky, *The Early Works of James Tenney*, P. Garland, Ed., Santa Fé: Soundings Press, 1984, pp. 193-203.
10. B. Gilmore, "James Tenney and the Poetics of Homage," *Contemporary Music Review*, 2008.
11. R. Miraglia, "Influences of phenomenology: James tenney's theory," *Axiomathes* 6, p. 273-308, 1995.
12. A. Prado, *Cartas celestes: uma uranografia sonora geradora de novos processos composicionais*, Universidade Estadual de Campinas, 1985.

Appendix

